

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER– VI (NEW) EXAMINATION – WINTER 2021****Subject Code:3161005****Date:26/11/2021****Subject Name:Fiber Optic Communication****Time:10:30 AM TO 01:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

MARKS

- | | | |
|------------|--|-----------|
| Q.1 | (a) Discuss the three attenuation window in fiber optic communication. | 03 |
| | (b) Define the term : (i) Quantum efficiency (ii) Cut off wavelength (iii) Numerical Aperture (iv) Responsivity | 04 |
| | (c) Write short notes on Synchronous optical fiber networks (SONET). | 07 |
| OR | | |
| Q.2 | (a) Give the comparison of S.M. and M.M. fibers. | 03 |
| | (b) “The optical power launched into a fiber does not depend on the wavelength of the source but only on it’s brightness” Justify | 04 |
| | (c) Derive the relationship between acceptance angle and numerical aperture with the use of ray theory. | 07 |
| OR | | |
| | (c) Derive the equation for the power launched from LED Source in to a G.I. fiber. | 07 |
| Q.3 | (a) List the different types of lensing schemes used in optical system. | 03 |
| | (b) Explain stimulated emission mechanism. | 04 |
| | (c) Explain Amplitude and shape degradation of optical signal in fiber. | 07 |
| OR | | |
| Q.3 | (a) Give the comparison between LED and LASER. | 03 |
| | (b) Classify the different losses in optical fiber. | 04 |
| | (c) A multimode step index fiber with a core diameter of 80 μm and a relative index difference of 1.5% is operating at a wavelength of 0.85 μm . If the core refractive index is 1.48, calculate:
(i) Normalized frequency of fiber
(ii) Total number of guided modes. | 07 |
| Q.4 | (a) Explain the fiber drawing process in brief. | 03 |
| | (b) Explain V groove fiber splicing technique. | 04 |
| | (c) Discuss the various dispersion mechanisms that contribute to pulse broadening during the fiber transmission. | 07 |
| OR | | |
| Q.4 | (a) Explain set up for the measurement of dispersion. | 03 |
| | (b) Explain double crucible method of fiber fabrication. | 04 |
| | (c) Discuss optical power loss model for a point to point link. | 07 |
| Q.5 | (a) What is population inversion? | 03 |
| | (b) Describe the architecture and amplification mechanism of Erbium Doped Fiber Amplifier. | 04 |
| | (c) Explain the basic principles of WDM and DWDM with respect to optical | 07 |

fiber system.

OR

- | | | | |
|------------|------------|--|-----------|
| Q.5 | (a) | Explain mechanical misalignment in fiber joining process. | 03 |
| | (b) | Write a technical note on optical receiver operation. | 04 |
| | (c) | Briefly explain the principal of optical detection. Discuss the major advantages of APD over P-i-n photodiode. | 07 |
